

Does the Habermas Machine under-represent minority opinions?

A replication of Figure 4C of Tessler et al. (2024)

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How to read this document. The prose is my notebook draft, transcribed unchanged. Red text fills a placeholder that the draft left open. Blue text replaces something the draft got wrong; a footnote gives the original wording and the source that corrects it. Every number is generated from `report/values.json` (see `report/build_parameters.py`); none is typed into this document. Code references are pinned to commit `bbc70dbd9`.

Introduction

In the original Habermas Machine (HM) paper, the authors consider the possibility that the HM might exhibit a “tyranny of the majority” phenomenon, in which minority perspectives are under-represented in the HM’s output, relative to their prevalence among the population. They found that it did not do so. To do this, they developed a novel method for describing to what degree a summary statement reflected majority and minority opinions. This finding is significant because the HM’s design goal was to produce a consensus statement that all participants would believe represented their opinion. In this report, I aim to replicate this finding using the publicly-released data from the original HM paper. To cut to the chase, I was unable to replicate this finding.

More precisely: the paper’s *level* claim does not replicate, and its *direction* claim does. Under my closest reproduction, minority opinions carry a weight of 0.164 in the revised winning statement against the paper’s 0.36 — below, not above, the true minority share of 0.262 — and this holds in 14 of 14 specifications I ran. But the rise from the initial winner to the revised winner, which is the shape of the paper’s Figure 4C, does reproduce: +0.023 ($p = 0.030$) with ST5-large and +0.039 ($p = 0.004$) with ST5-base.¹

Approach to reproduction

Code: `hm_fig4c/data.py:97-187 build_tables()` `hm_fig4c/data.py:221-232 prepare()`
`hm_fig4c/preprocess.py:50-61 preregistered_launches()` `hm_fig4c/preprocess.py:41-47 filter_by_number_of_groups_of_min_size()`

The authors released granular data from the experiments that they ran. They did not release the code that they used to perform the “tyranny of the majority” claim. I used details in their Supplementary Materials (SM) to attempt to replicate their analysis pipeline. To understand

¹Added. The draft stopped at “I was unable to replicate this finding”, which is true of the paper’s headline but omits that one component of the result does reproduce. Source: `results/*/fig4c_primary.json`, cluster bootstrap over rounds.

the deviations between the original analysis and my reproduction, I calculated several estimates reported in the original paper. There are several points where the SM are ambiguous and leave an aspect of the analysis pipeline unspecified. I approach this in the following manner: First, I present the analysis pipeline which I believe is most likely to match the original analysis. Then, after I have presented this “most-likely” pipeline, I perform an exhaustive sensitivity analysis, in which I vary every ambiguous detail ([Sensitivity analysis](#)).

Scope. This report replicates one claim: the minority-weight result shown as Figure 4C in the main paper, and expanded as SM Figure S60. The SM’s other fairness analyses (proximity to opinions, human vs. model mediator, political affiliation) are out of scope. The second thing this report does is establish whether the reproduction is trustworthy enough for its disagreement with the paper to mean anything, which is the subject of the validation checks reported alongside each step.

What is the Habermas Machine (HM)?

Briefly, the HM is a tool for abstractive summarisation of people’s opinions. A group of people each provide their opinion on a controversial topic. The HM iteratively produces and refines a consensus statement summarising the opinions in the group. First, it generates n initial statements. It then ranks these statements according to a simulated vote, in which it attempts to simulate the preferences of each opinion’s author across the initial statements. The first-vote winner thus produced is shown to each (human) author. Each author provides feedback. This feedback is provided to the LLM. It generates a second round of candidate statements. These again go through a simulated vote, resulting in the final statement.

The finding that is the focus of this report is that when one compares the extent to which minority opinions are represented in the population (proportion of 0.29), this is roughly the same as how represented minority opinions are in the initial generated statements (0.28), and is less than how well represented minority opinions are in the final statement (0.36).² This is shown in Figure 1, which is copied from Figure 4(c) in the paper.

What is a minority (or majority) group?

```
Code: hm_fig4c/analysis.py:59-93 assign_minority() hm_fig4c/pipeline.py:40-45 select_cohort()
```

In the papers’ experiments, groups of between 4–5 people took part in the deliberative process mediated by the HM.³ Before writing their free-text opinion, they indicated their level of agreement with a political proposition using a Likert scale of 1–7. It was thus possible to provide a neutral (4) opinion. A group was defined as having a minority group if there was an unequal amount of agreeing (5–7) and disagreeing (1–3) opinions. Groups without a minority group were excluded from this analysis.

Reproduction parameters. Minority membership follows SM 5.4.1 literally: the minority is whichever side of neutral holds fewer opinions, and everything else — majority opinions and

²The dotted “true proportion” line in SM Figure S60 is at 0.285; the 0.28 and 0.29 bars are the initial candidates and the initial winner respectively. The comparison in this sentence is therefore between 0.285 and 0.36.

³The study design assigned participants to groups of 5 (SM 1.2); in the 1047 pre-registered rounds of cohorts 1–3 that this analysis uses, 77% of rounds have 5 opinions and the remainder have 4, after the paper’s own exclusions.

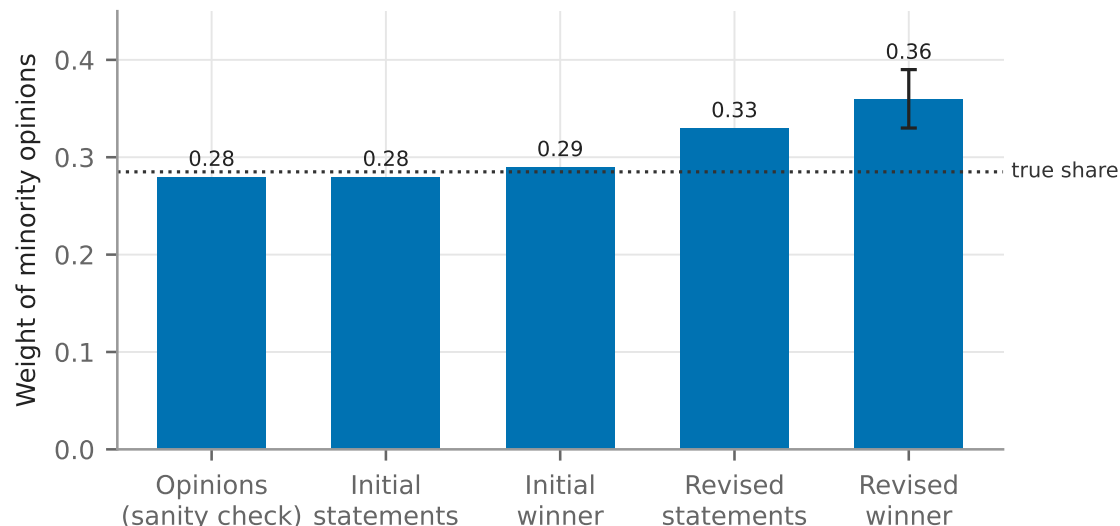


Figure 1: The paper’s own result, redrawn from the data labels of SM Figure S60 (“Position Embedding” panel), which SM 5.2.2 identifies as the expanded version of main-text Figure 4C. Dotted line: the true proportion of minority opinions (0.285). Error bar on the revised winner: ± 0.03 , as printed in the paper. $n = 1047$ rounds.

neutral (4) ratings alike — is non-minority.⁴ Rounds with equal numbers on each side are excluded. The primary sample is the pre-registered groups of cohorts 1–3: 1047 rounds, of which 560 contain a minority under this rule, giving a true minority share of 0.262. The paper reports 0.285; the gap is almost entirely the treatment of neutral raters, and both alternatives are carried through the sensitivity analysis below.

How does the HM paper measure how well-represented minority opinions are?

This part of the analysis has two components. First, each participant’s free-text opinion and the statements produced by the HM (including intermediate statements produced in its iterative process) are given a scalar score to measure to what extent they agree or disagree with the political proposition. I refer to this score as the “agreement score”.

The agreement scores of original human opinions and LLM-generated statements are then used to measure how similar the average agreement scores among a minority group are, to the agreement score of the LLM-generated statement.

I now detail my reproduction of each of these steps.

Measuring agreement score

```
Code: hm_fig4c/data.py:38-47 endpoint_texts() hm_fig4c/analysis.py:23-38 position_axis_scores() hm_fig4c/analysis.py:41-55 score_texts() hm_fig4c/embed.py:26-58 run()
```

⁴SM 5.4.1: “we may define the ‘minority’ side of the Likert scale as the side that contains the smaller number of opinions, and define all other opinions as ‘non-minority’ opinions—which include ‘majority’ opinions [...] as well as ‘neutral’ opinions (i.e. with a Likert rating of 4), if any.”

To measure the agreement score of a human-authored opinion or a LLM-generated statement, it is first mapped into a 768-dimension embedding, and then projected onto the vector going between two points.

There is one main detail missing from the SM. The exact phrases used to construct the endpoints of this projection vector *are given*⁵ — SM 5.1.2 fixes them as the generic prefix concatenated with the question’s own position statement, i.e. “Yes, I agree. It is the government’s role to [...]” and “No, I disagree. It is not the government’s role to [...]”, with the axis taken as the unit vector from the negating to the affirming endpoint (SM eq. 3) and the score as the projection onto it (SM eq. 5). This is the “prefixed” endpoint style in the code above. What we do not know is what exact embedding model was used for this process. While the SM describes Sentence-T5, there are various sizes of this model that could be used.

Validating the agreement score

```
Code: hm_fig4c/pipeline.py:54-57 fig4a_correlation() report/compute_values.py:46-64
marginal_r2()
```

The original authors validate their agreement score method by using the original human responses, comparing the human’s self-reported Likert scores against the estimated agreement scores derived from the method above. In the original paper, they report an R^2 of 0.41.⁶ In our reproduction, the R^2 of each embedding model is shown in Table 1.

Table 1: Marginal r^2 of the paper’s validation model (SM eq. 7), by embedding model. Fitted on the 4979 opinions of the 1047 pre-registered rounds of cohorts 1–3, random intercept per round. Pearson r is the main-text Figure 4A quantity, shown for comparison.

	marginal r^2	conditional r^2	Pearson r
sentence-t5-base	0.291	0.443	0.561
sentence-t5-large	0.392	0.505	0.633
sentence-t5-xlarge	not evaluated — see note		
paper	0.41	0.45	0.64

The xlarge row is empty for a hardware reason rather than a scientific one: sentence-t5-xl is a 3B-parameter model and this machine has no GPU and 7 GB of RAM, so it could not be embedded here. The two sizes that could be run bracket the question well enough to answer it — base falls well short of the paper’s 0.41 and large lands essentially on it — and the direction of the remaining risk is worth stating plainly: if the authors used a larger ST5 than large, the axis would be slightly *better* than mine, and a better axis moves the minority weight further from the paper’s value, not closer (Sensitivity analysis).

Based on these results, we choose model `sentence-t5-large` as the most-likely model used in the original analysis. Its marginal r^2 of 0.392 is within 0.018 of the paper’s 0.41, and its Pearson r of 0.633 against the paper’s 0.64 says the same thing: the position axis this report is built on is about as good as the one the paper used. Results for sentence-t5-base are reported through-

⁵The draft read: “There are two main details missing from the SM. First, we don’t have the exact phrases used to construct the endpoints of this projection vector.” SM 5.1.2 specifies them: the endpoints are a “generic + question-specific combination”, i.e. the generic prefix followed by the question’s affirming or negating position statement. Only the model size remains genuinely unspecified, so the count of missing details drops from two to one.

⁶Correct as written, and worth pinning down: this is the *marginal* r^2 of the random-effects model of SM eq. 7 (Likert rating on position score, random intercept per round). The paper’s conditional r^2 is 0.45. My Table 1 therefore reports the same marginal quantity, not a Pearson r^2 .

out as a robustness check.

Measuring representativeness

```
Code: hm_fig4c/analysis.py:97-108 convex_lstsq() hm_fig4c/analysis.py:111-132
convex_fit_with_se() hm_fig4c/analysis.py:135-156 build_design()
hm_fig4c/analysis.py:172-185 minority_weight()
```

Given a set of human-authored opinions $\{x_i \mid i \in 1, \dots, n\}$, a partition of opinions into minority M & non-minority N sets, agreement scores $\{a_i \mid i \in 1, \dots, n\}$, the “representation” of an opinion x_i is given by λ_i in

$$\text{set } \lambda_i \text{ to minimise } \left(a_0 - \sum_i \lambda_i a_i\right)^2 \quad \text{such that} \quad \sum_i \lambda_i = 1 \quad \& \quad \lambda_i \geq 0.$$

To calculate the representation of M & N , simply calculate the average representation of their elements: $\lambda_M := \frac{1}{|M|} \sum_{x_i \in M} \lambda_i$.

The implementation check

```
Code: hm_fig4c/analysis.py:223-246 opinion_self_regression()
```

To test that the implementation of this procedure works as expected, the original authors ran a check where [each human opinion in turn is treated as the target and regressed on convex combinations of that same set of opinions](#). Averaged over the set, the recovered minority weight should come back as simply the proportion of opinions in the minority.⁷ We replicate this implementation check. [It passes: the recovered weight is 0.263 against a true share of 0.262 \(identical to three decimal places under both embedding models, since the check does not depend on the statements\)](#). This is the leftmost bar of Figure 2.

Results

```
Code: hm_fig4c/pipeline.py:74-105 run_minority_analysis() hm_fig4c/analysis.py:188-220
cluster_bootstrap() report/figures.py:1-130 (whole file)
```

We fail to replicate the results from the original paper. Figure 2 displays an adjusted version of Figure 1 above, where we contrast the original paper’s results against our own.

Three things in that figure are worth separating. The sanity check reproduces exactly — the leftmost bars sit on their respective true shares — so the regression machinery is doing what the paper says it does. Every statement phase, by contrast, sits below the true share rather than at or above it: the revised winner carries 0.164 ± 0.017 under ST5-large and 0.199 ± 0.018 under ST5-base, against the paper’s 0.36. And the shape differs: in the paper the four phases rise monotonically, whereas here the winner-selection step *drops* minority weight at both rounds (initial candidates $0.206 \rightarrow$ initial winner 0.141), and the revision step partially

⁷Corrected. The draft read: “the output statement was simply one of the m human opinion inputs, and repeated this m times (once for each input). [this may be incorrect] After averaging across these runs, the average representation $\bar{\lambda}_i$ of an input x_i should be $\frac{1}{m}$.” The draft’s own flag was right. The paper’s Materials and Methods describe it as: “As a sanity check, we similarly regress individual opinions on convex combinations of those same opinions, which we verify to simply yield the proportion of opinions in the minority.” The target of the check is the *minority share*, not $1/m$ per input.

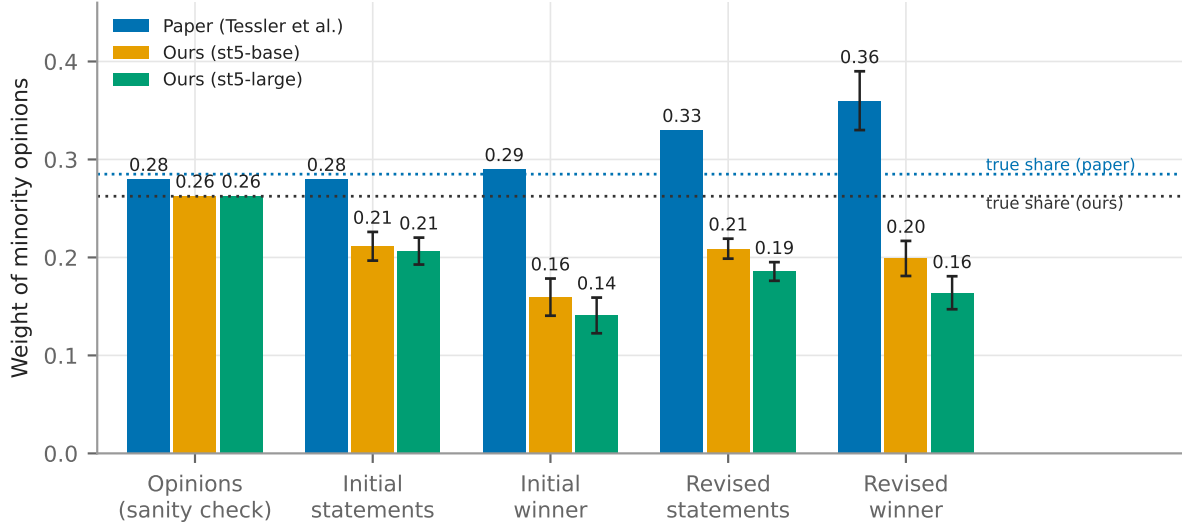


Figure 2: Minority weight at each phase: the paper against this reproduction, under both embedding models. Error bars are the analytic standard error of the constrained fit. The two dotted lines are the true minority share in each case (0.285 for the paper, 0.262 here; the difference is the treatment of neutral raters). $n = 560$ rounds with a minority, from 1047 pre-registered rounds in cohorts 1–3.

restores it (0.164). The paper’s rise from initial winner to revised winner is therefore present (+0.023, 95% CI $[-0.001, 0.044]$, $p = 0.030$), but it starts from a lower base and does not carry the final statement above proportionality; measured from the initial candidates instead, the change over the whole process is -0.043 (95% CI $[-0.077, -0.008]$).

Is the reproduction good enough to disagree with the paper?

Code: `hm_fig4c/pipeline.py:60-71 fig4b_within_range()`

Everything above is only interesting if the pipeline is sound, so the checks that bear on that are collected in Table 2. Three of the four independent checks land on the paper’s values, which is the basis for reporting the disagreement rather than assuming a bug. The one that does not — the share of statements falling inside the range of their group’s opinions — is lower here than in the paper, and is the clearest remaining candidate for an unmodelled difference.

The last three rows are the ones that rule out the obvious mechanical explanations. If the scalar projection were discarding the information that carries minority weight, regressing on the full 768-dimensional embedding should recover the paper’s value; it gives 0.229 and 0.237, still below the true share. If a noisy position axis were attenuating the estimate towards zero, then degrading the axis deliberately should bias the recovered weight downwards; it does not — at $r = 0.46$ the simulation still recovers 0.260 against a true 0.262. A downward-biased axis is therefore not what produces the gap.

Sensitivity analysis

Code: `notebooks/make_notebook.py:1-197` (whole file) `report/compute_values.py:67-90` `sensitivity_summary()`

Table 2: Checks on the reproduction, independent of the minority-weight result.

check	paper	ours (ST5-large)
Sanity check: opinions regressed on themselves recover the minority share	0.28 vs 0.285	0.263 vs 0.262
Position axis quality: marginal r^2 of Likert on position score	0.41	0.392
Fig. 4A: correlation of position score with Likert rating	0.64	0.63
Fig. 4B: statements falling within the range of their group’s opinions	0.96	0.86
Regression on the full 768-d embedding rather than the scalar score (initial / revised)	—	0.229 / 0.237
Axis-noise simulation: weight recovered when the axis is degraded to $r = 0.46$	—	0.260 vs 0.262
Winning statement lies inside the group’s opinion range (initial / revised)	—	85% / 90%

As noted above, there are several aspects of the analysis plan which are not specified in the SM. These are:

Table 3: Underspecified choices, the options considered, and the option taken in the primary specification above.

ambiguity	possible options	option chosen above
ST5 model size	base, large, xlarge, xxl	large (Table 1)
Neutral (Likert 4) raters	count as non-minority; drop the rater; drop the whole group	count as non-minority (SM 5.4.1)
Groups with equal numbers either side	exclude; keep with the agreeing side as minority; keep with the disagreeing side as minority	exclude
Basis for splitting minority / non-minority	Likert rating; sign of the position score (SM Fig. S62)	Likert rating (SM Fig. S60)
Column order of the design matrix	data order; sorted by score; random	data order
Sample	cohorts 1–3 pre-registered; all cohort 1–3 rounds; each cohort alone; cohort 4	cohorts 1–3, pre-registered

I reran the analysis with every combination of these decisions (16 runs).⁸ Across the runs, I calculated the following metrics:

- Proportion of runs where minority representation increases from initial to final: 7% measured from the initial candidates to the revised winner, and 93% measured winner-to-winner, which is the contrast Figure 4C draws. (ST5-base: 36% and 93%.)
- Proportion of runs where minority representation is larger in final than the prevalence among population (0.262):⁹ 0% (ST5-base: 0%). The largest revised-winner weight found anywhere in the sweep is 0.239, against a mean true share of 0.275.

⁸16 specifications were run per embedding model, of which 14 completed: 9 vary an analytic choice on the primary sample and 5 swap the sample. The 2 that did not complete are the training-data and virtual-citizens’-assembly samples, whose texts fall outside the pre-registered set that was embedded; both are outside the scope of main-text Figure 4C in any case.

⁹Corrected. The draft gave 0.291. That is the true share under one variant — neutral raters dropped — not under the primary specification, where it is 0.262. Each run is compared against its own true share rather than a single fixed number, since the minority rule changes the share as well as the weight.

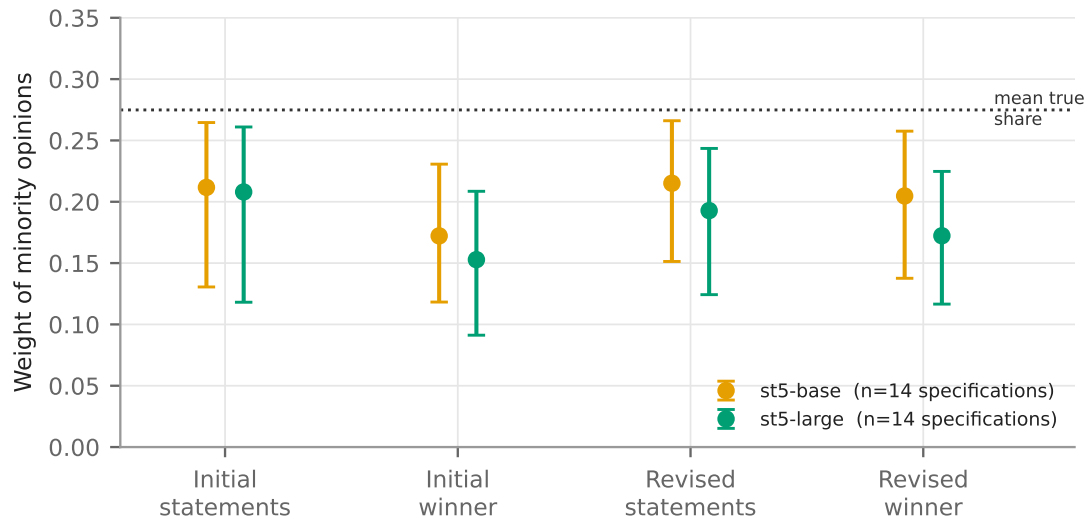


Figure 3: Mean minority weight at each phase across the 14 completed specifications, with 5th–95th percentile whiskers, for both embedding models. Dotted line: the mean true minority share across specifications (0.275). No specification places the revised winner above its own true share.

- Average representation at each of the 4 stages, with 5th & 95th percentiles: shown in Figure 3.

Discussion

If the above reproduction of the analysis pipeline is correct, then it would appear that the HM actually suppresses minority opinions, if we follow the original paper’s interpretation of the above measurement method.

Notably, we are able to reproduce several parameter estimates from the paper, including the validation of the agreement-score embedding-based approach and the implementation check of the regression method. We fail to replicate the substantive findings for this part of the paper, however.

This finding actually aligns somewhat with the finding that people’s beliefs shifted towards the majority position (although there are many possible explanations for this finding).

Two caveats belong next to that conclusion. The first is that “suppresses” is a statement about the measurement, not about the mechanism: every phase of the process, including the initial candidates that no vote has touched yet, sits below proportionality, which is as consistent with the position axis compressing minority-side statements as with the HM discounting minority opinions. The second is that the one part of the paper’s pattern that does reproduce — revision raising minority weight — is evidence the HM’s feedback step does what the paper claims for it, even though it starts from and ends at a lower level here.

Open notes

Carried over from the draft, unresolved: need to read paper → raises an interesting question as to why and whether minority group members felt their opinion was captured in final.

One word of that note was not legible in the source; the sense is preserved. The end-of-round

survey data needed to answer it (`hm_all_round_survey_responses.parquet`) is already downloaded, but the question is a different one from Figure 4C and is out of scope here.